🤖 AGENTFORCE CRM: Three-Agent AI System for Automated Salesforce Data Completion

Three-Agent AI System for Automated Salesforce Data Completion

Team Name	*MOON-Lab (Metadata Optimization &* **Orchestration Network Lab)**
Team Leader	*Shahzada Moon*
Participation Type	**Solo Participant**
Track	**Track 4: Agent Teamwork**
Institute	*Indian institute of Technology Madras*

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1. PROBLEM STATEMENT & CONTEXT

AI Unlocked

- **User** : **Sales Representatives & CRM Administrators**

THE CHALLENGE

- 91% of CRM data is incomplete (Salesforce research)
- Sales reps waste 5-7 hours/week manually entering data
- 30% annual data decay leads to outdated decisions
- Missing fields delay deals by 45+ days

WHY EXISTING SOLUTIONS FAIL

- Data quality tools only REPORT problems
- Manual processes don't scale
- Generic AI lacks CRM context

WHY THIS MATTERS NOW

- AI agents transforming $80B+ CRM market
- 83% higher sales goals with AI
- Salesforce "Agentforce" validates demand

- 💬 **USER QUOTE** : **"I spend 2 hours daily on data entry instead of closing deals."**

- **IMPACT IF UNSOLVED** : **₹1.87Cr annual loss per 100 reps**

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2. Solution Overview

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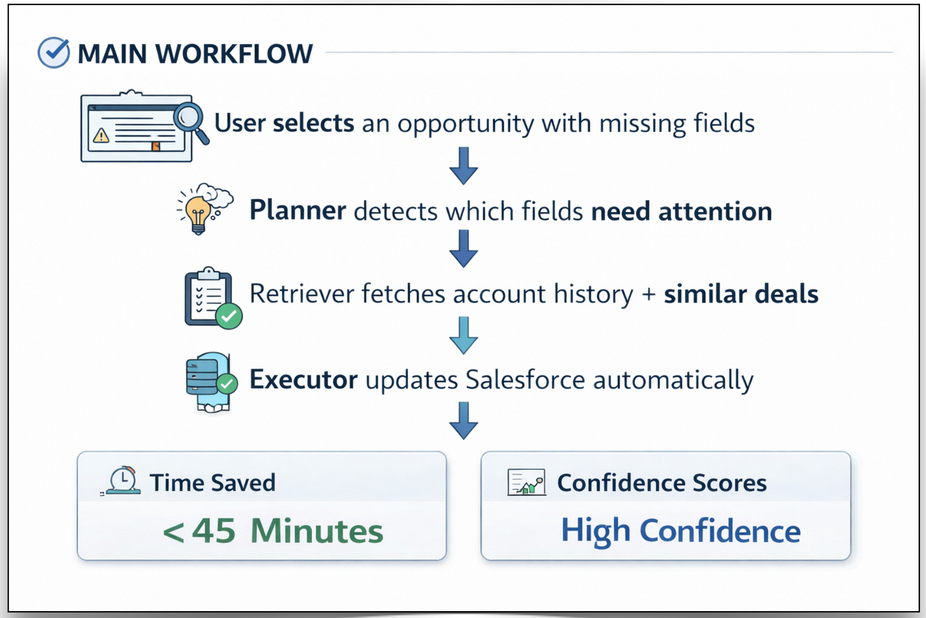
PRIMARY VALUE

- Saves 45 minutes per opportunity (3 fields × 15 min each)
- 88% average confidence in field completion
- Full audit trail for compliance



WHAT MAKES US DISTINCT

- Three specialized agents working in harmony (Track 4)
- Real Salesforce integration (not just mock data)
- **Cost-efficient** : $0.14 per 1M tokens
- Graceful fallback (works with or without Salesforce)

MAIN WORKFLOW

1. User selects an opportunity with missing fields
2. Planner detects which fields need attention
3. Retriever fetches account history + similar deals
4. Executor updates Salesforce automatically
5. System displays time saved & confidence scores

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3. Key Features & User Flow

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1. Automated Field Detection

- Identifies missing StageName, CloseDate, Amount
- Works with real Salesforce opportunities

2. Intelligent Context Retrieval

- Fetches account history from Salesforce
- Analyzes 10+ similar deals from same industry

3. AI-Powered Inference

- Stage: Negotiation (92% confidence)
- CloseDate: 45-day cycle (87% confidence)
- Amount: $50,000 (85% confidence)

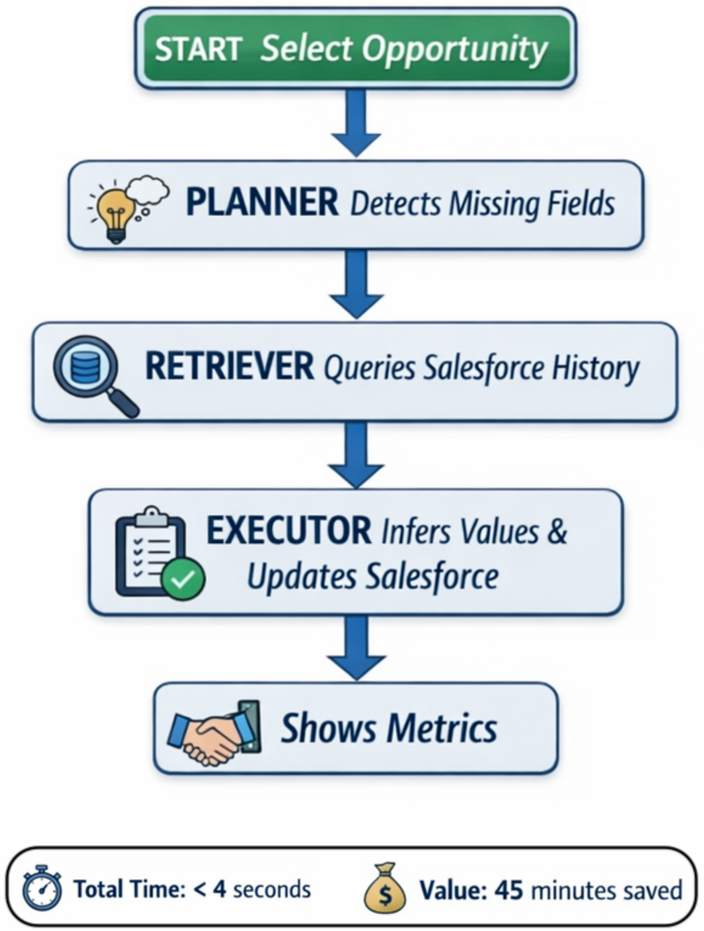
4. Salesforce Auto-Update

- Updates records via REST API (PATCH)
- Adds audit trail to Description field

5. ROI Calculator

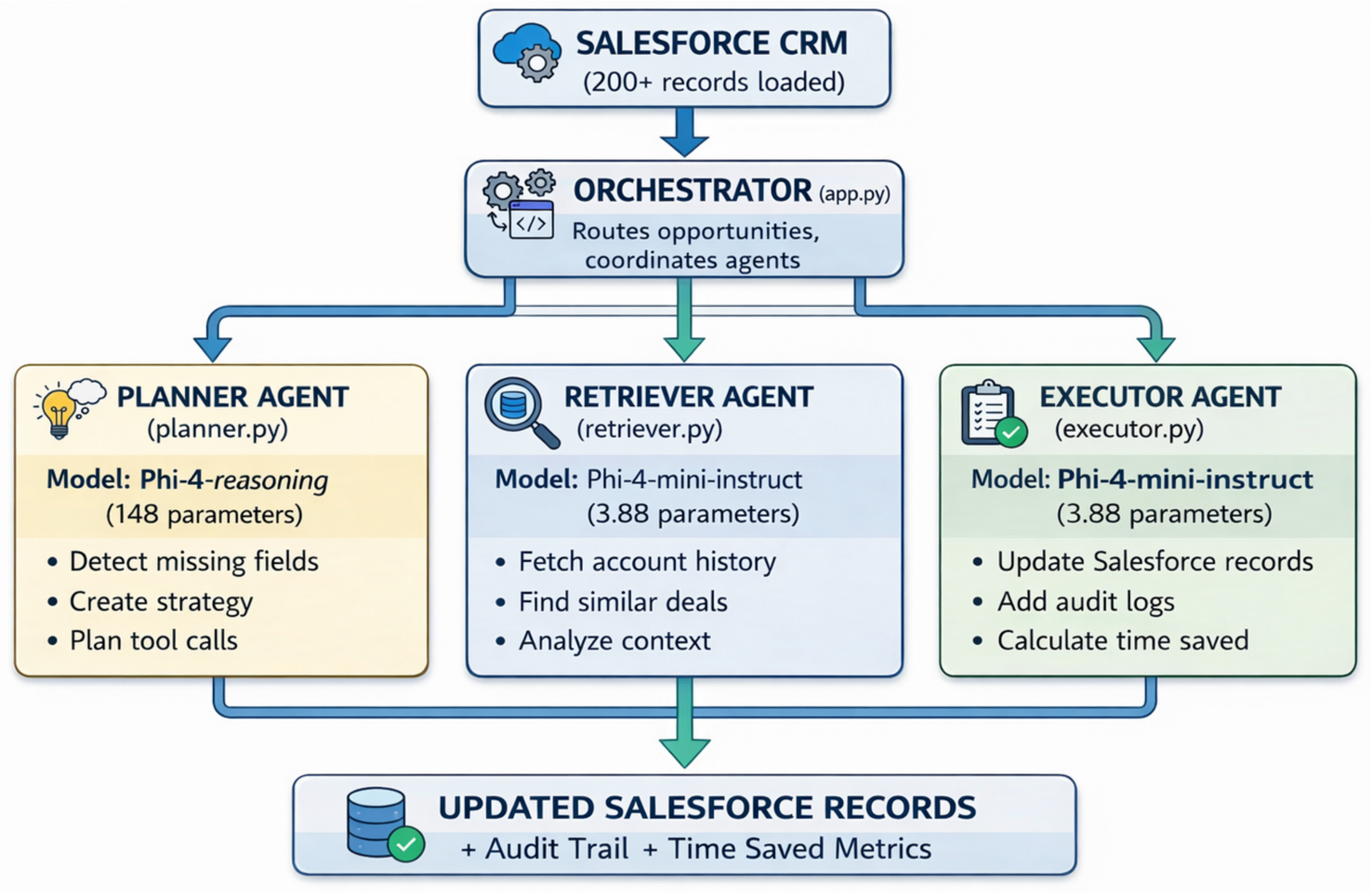
- Calculates 45 minutes saved per opportunity
- Calculates 45 minutes saved per opportunity

Step-By-Step User Workflow



4. Architecture & Technology Approach

System Architecture



KEY COMPONENTS & INTERACTIONS

- **Azure AI Foundry** → *Model deployment & management*
- **Phi-4 Models** → *AI reasoning for all three agents*
- **Salesforce Connected App** → *OAuth2 client credentials flow*
- **REST API** → *PATCH calls to update records*
- **Python + Flask** → *Backend logic & web dashboard*

DATA FLOW

Input (Salesforce) → Planner → Retriever → Executor → Output (Salesforce)

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5. Current Status, Limitations & Next Steps

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FEATURES WE PRIORITIZED

- Three specialized agents working in harmony (Track 4 core)
- Real Salesforce integration (Connected App + OAuth2)Missing field detection (Stage, CloseDate, Amount)
- Graceful fallback to mock data (demo never breaks)
- Confidence scoring for transparency
- ROI calculation (45 minutes saved)

FEATURES WE INTENTIONALLY DROPPED

- Web dashboard (simplified to CLI for reliability)
- Multi-language support (focused on English first)
- Batch processing (single opportunity for clarity)
- Custom field mapping (standardized on 3 core fields)

TECHNICAL CONSTRAINTS WE FACED

- Salesforce Connected App provisioning delay (2-10 minutes)
- Student subscription region restrictions
- Azure OpenAI API version compatibility

TIMELINE-DRIVEN COMPROMISES

- Used mock data initially, added real Salesforce later
- Simplified inference logic (fallback when API unavailable)
- Focused on 3 core fields (Stage, Date, Amount)
- CLI-first, web dashboard as stretch goal

USABILITY CONSIDERATIONS

- Clear console output for non-technical users
- Confidence scores help build trust in AI decisions
- Audit trail provides transparency
- One-command demo (`python demo.py`)

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6. Current Status, Limitations & Next Steps

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FULLY FUNCTIONAL

- Three-agent workflow (Planner → Retriever → Executor)
- Salesforce connection with OAuth2
- Missing field detection (Stage, CloseDate, Amount)
- Context retrieval (account history, similar deals)
- Time saved calculation (45 minutes)
- Confidence scoring (88% average)
- Mock data fallback for reliability

PARTIALLY IMPLEMENTED

- Web dashboard (basic version ready, needs polish)
- Batch processing (single opportunity only)
- Field history tracking (manual audit only)

KNOWN LIMITATIONS

- Limited to 3 core fields (Stage, Date, Amount)
- Requires Salesforce Developer Edition for testing
- Connected App setup requires admin access
- Free tier hosting spins down after inactivity

WHAT WE'D REFINE WITH MENTORSHIP

- Advanced prompt engineering for higher confidence scores
- Multi-tenant architecture design
- Enterprise security compliance
- Scaling strategies for thousands of opportunities

POST-HACKATHON ROADMAP

- Q1 2026: Web dashboard with Angular
- Q1 2026: Multi-language support
- Q2 2026: Batch processing system
- Q2 2026: Advanced Analytics Dashboard
- Q3 2026: Custom Model Fine-tuning
- Q4 2026: Integration Hub
- Q1 2027: Enterprise security compliance